A TO Z MASTER GUIDE

Excel for Data Analysis

Formulas to Dashboards - Power Query - PivotTables - Interview Questions

Prepared specially for Koushik
Beginner-friendly A to Z edition

Inside: FORMULAS | LOOKUPS | CLEANING | PIVOTS | POWER QUERY | DASHBOARD

How to use	Oka topic chadivi code ni own-ga type cheyyi. First output guess cheyyi, tarvatha run cheyyi. Interview answer boxes ni loud-ga 2 times practice cheyyi.

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How to Use This Guide

Ee PDF Excel ni zero nunchi interview level varaku nerchukovadaniki. Prathi main topic lo step, reason, expected result, common mistake and interview answer untayi. Menu names software version batti konchem change avvachu, kani workflow same untundi.

Step	What you do	Why
1. Observe	Concept and example chaduvu	Meaning ardham avutundi
2. Predict	Result mundu guess cheyyi	Logic build avutundi
3. Perform	Own-ga steps repeat cheyyi	Tool memory vastundi
4. Explain	Result enduku vachindo cheppu	Interview clarity vastundi
5. Practice	Two new questions solve cheyyi	Independent skill build avutundi

Interview answer formula	Definition one line -> process 2 or 3 points -> small example -> business use -> limitation. Ee order lo answer chepthe neat-ga and confident-ga untundi.

1. Excel Foundation

1.1 Workbook, worksheet, cell and range

Term	Meaning	Example
Workbook	Excel file containing sheets	Sales.xlsx
Worksheet	One grid inside workbook	Orders sheet
Cell	Intersection of row and column	B5
Range	Group of cells	A2:D100
Table	Structured dynamic data range	OrdersTable

1.2 Interface path

- Ribbon: Home, Insert, Page Layout, Formulas, Data, Review, View tabs.
- Name Box shows active cell/range; Formula Bar shows actual value or formula.
- Status Bar can show Sum, Average and Count for selected numeric cells.
- Use Ctrl+F1 to collapse/expand ribbon; View > Freeze Panes to keep headers visible.

Easy Telugu	Excel lo data enter chesi, formulas tho calculate chesi, PivotTable/charts tho summary create chestam. Data neat table format lo unte analysis easy.

1.3 Data types and number formats

Excel stores underlying values and displays them through formats. A date is usually a serial number with a date format; percentage 25% is stored as 0.25.

Value	Underlying idea	Format result
0.25	Decimal number	25%
45292	Date serial	01-Jan-2024 (depending on system)

Value	Underlying idea	Format result
1250.5	Number	1,250.50 or currency
'00125	Text forced by apostrophe	00125

Why total is wrong?	Number text-ga store ayithe SUM ignore cheyyachu. Green warning, left alignment, VALUE/NUMBERVALUE or Power Query data type tho convert cheyyi.

2. Formula Basics and Cell References

2.1 Formula evaluation

```
=B2*C2
=(B2*C2)*(1-D2)
=SUM(E2:E100)
```

Every formula starts with `=`. Excel first referenced cell values reads, operators/functions evaluate chestundi, final value display chestundi.

2.2 Relative, absolute and mixed reference

Reference	Copy behavior	Use
A1	Row and column both move	Normal row-wise calculations
$A$1	Neither moves	Fixed tax/target cell
$A1	Column fixed, row moves	Matrix by fixed field column
A$1	Row fixed, column moves	Headers/rates across columns

```
Price in B2, Quantity in C2, Tax rate in H1
=B2*C2*(1+$H$1)
```

Result enduku?	Formula down copy chesthe B2/C2 row numbers move avutayi. $H$1 fixed kabatti every row same tax rate use chestundi. F4 press chesi reference types cycle cheyyachu.

2.3 Operator order

Parentheses, exponent, multiplication/division, addition/subtraction order apply avutundi. Business formula intention clear-ga undadaniki parentheses use cheyyi.

3. Essential Functions

3.1 Math and statistics

Need	Formula
Total	=SUM(E2:E100)
Average	=AVERAGE(E2:E100)
Count numeric	=COUNT(E2:E100)
Count nonblank	=COUNTA(A2:A100)
Min / Max	=MIN(E2:E100), =MAX(E2:E100)
Rounded value	=ROUND(E2,2)
Rank	=RANK.EQ(E2,$E$2:$E$100,0)

3.2 IF, IFS, AND, OR and IFERROR

```
=IF(E2>=100000,"High","Normal")
=IF(AND(E2>=50000,F2>0),"Good","Review")
=IFS(E2>=100000,"A",E2>=50000,"B",TRUE,"C")
=IFERROR(A2/B2,0)
```

IF flow	Logical test first. TRUE ayithe second argument, FALSE ayithe third argument. Nested IF kanna many bands ki IFS/readable lookup table better.

3.3 Conditional aggregation

```
=SUMIF(B:B,"South",E:E)
=SUMIFS(E:E,B:B,"South",C:C,">="&DATE(2026,1,1))
=COUNTIFS(B:B,"South",F:F,"Profit")
=AVERAGEIFS(E:E,B:B,"South",D:D,"Technology")
```

Function	Arguments idea
SUMIF	One condition + sum range
SUMIFS	Sum range first, then multiple condition range/value pairs
COUNTIFS	Multiple condition pairs
AVERAGEIFS	Average range first, then condition pairs

Common mistake	SUMIFS ranges same size undali. Criteria operator cell value tho combine chesthe `">="&H1;` laga & use cheyyi.

4. Lookup Functions

4.1 XLOOKUP - recommended modern lookup

```
=XLOOKUP(A2,Products[Product ID],Products[Product Name],"Not found")
```

Argument	Meaning
A2	Value to find
Products[Product ID]	Lookup column
Products[Product Name]	Return column
Not found	Fallback when no match

How result came	A2 ID ni Product ID column lo exact-ga search chesi same row Product Name return chestundi. Return column left/right ekkada unna problem ledu.

4.2 VLOOKUP and exact match

```
=VLOOKUP(A2,$H$2:$K$100,3,FALSE)
```

VLOOKUP first column lo searches and specified column number returns. Exact lookup ki FALSE/0 use cheyyi. Columns insert chesthe hard-coded index break avvachu.

4.3 INDEX + MATCH

```
=INDEX(K2:K100,MATCH(A2,H2:H100,0))
```

MATCH position finds; INDEX that position nunchi value returns. Flexible but XLOOKUP available ayithe simpler.

4.4 Approximate lookup

```
=XLOOKUP(E2,$H$2:$H$6,$I$2:$I$6,"",-1)
```

Important	Approximate match grading/slabs ki boundary table sorted and logic clear undali. Exact product/customer IDs ki approximate match use cheyyaku.

5. Text and Date Functions

5.1 Text cleaning

Need	Formula
Remove extra spaces	=TRIM(A2)
Remove non-printing chars	=CLEAN(A2)
Case	=UPPER(A2), =LOWER(A2), =PROPER(A2)
Combine	=TEXTJOIN(" ",TRUE,A2:C2)
Extract	=LEFT(A2,3), =RIGHT(A2,4), =MID(A2,2,5)
Find position	=SEARCH("-",A2)
Replace	=SUBSTITUTE(A2,"old","new")

5.2 Dates

```
=DATE(2026,8,31)
=YEAR(A2)
=MONTH(A2)
=TEXT(A2,"mmm-yyyy")
=EOMONTH(A2,0)
=NETWORKDAYS(A2,B2,Holidays[Date])
=DATEDIF(A2,B2,"y")
```

Sorting trap	Month name TEXT result text kabatti alphabetic sort avvachu. Actual date/month number helper column maintain chesi display format use cheyyi.

6. Dynamic Arrays

6.1 FILTER, SORT and UNIQUE

```
=FILTER(Orders,Orders[Region]="South","No rows")
=SORT(FILTER(Orders,Orders[Sales]>1000),5,-1)
=UNIQUE(Orders[Customer Name])
=SORT(UNIQUE(Orders[Region]))
```

These formulas return multiple cells as a spill range. Place formula where surrounding cells are empty.

6.2 Spill error

#SPILL!	Output area lo existing value/merged cells/table boundary block chesthe spill error. Blocked cells clear cheyyi or formula location change cheyyi.

7. Data Cleaning and Validation

7.1 Convert range to Excel Table

Step	Action	Why/result
1	Select data -> Ctrl+T	Create dynamic table
2	Confirm My table has headers	Headers become field names
3	Table Design -> Table Name	Readable structured references
4	Add new row	Formulas/format expand automatically

7.2 Cleaning tools

- Data > Remove Duplicates: select columns defining a duplicate; keep backup first.
- Data > Text to Columns: split delimiter/fixed width; ensure empty destination columns.
- Home > Find & Select > Replace: bulk corrections; use carefully.
- Flash Fill (Ctrl+E): infer pattern from examples; verify all rows.
- Data > Data Validation: list, number/date rules and input/error messages.
- Go To Special > Blanks: inspect missing cells before filling.

Duplicate reason	Whole row duplicate and business duplicate different. Same customer name two genuine people kavachu. Unique business key decide chesi remove cheyyi.

8. Sort, Filter and Conditional Formatting

8.1 Multi-level sort

Select table -> Data > Sort -> Add Level. Example: Region A-Z, then Sales Largest to Smallest. Header option verify cheyyi, lekapothe header data lo mix avvachu.

8.2 Filter

Ctrl+Shift+L toggles filters. Text/Number/Date Filters use cheyyi. Filtering rows hide chestundi; data delete kaadu.

8.3 Conditional formatting

Need	Path	Reason
Highlight losses	Home > Conditional Formatting > Less Than 0	Risk scan
Top values	Top/Bottom Rules	Priority
Heatmap	Color Scales	Pattern visibility
Custom row rule	New Rule > Use a formula	Multi-column logic

```
Custom rule applied to A2:F100
=$F2<0
```

Why $F2?	Column F fixed kabatti every formatted cell same row Profit ni checks. Row relative kabatti each row own Profit check chestundi.

9. Charts - Which Visual and Why

Question	Best starting visual	Avoid
Compare categories	Bar/column	Too many pie slices
Trend over time	Line	Unsorted date categories
Part of whole	100% stacked or simple pie	Many categories
Relationship	Scatter	Treating correlation as causation
Distribution	Histogram/box plot where available	Only average
Actual vs target	Bar/column with target line	3D charts

9.1 Create a chart

Step	Action	Expected result
1	Select clean summary/table	Correct source range
2	Insert > Recommended Charts or chart type	Visual created
3	Chart Design > Select Data	Verify series/categories
4	Add title, labels only when useful	Readable message
5	Remove clutter and 3D effects	Honest comparison

10. PivotTables - Step by Step

10.1 First PivotTable

Step	Click/action	Why/result
1	Click inside Orders table	Excel detects source
2	Insert > PivotTable	Choose new/existing sheet
3	Drag Region to Rows	One row per region
4	Drag Sales to Values	Default Sum if numeric
5	Drag Category to Columns	Split totals by category
6	Value Field Settings > Number Format	Readable currency
7	Right-click > Refresh	Include source changes

Why Count instead of Sum?	Sales column text-ga/mixed data ayithe PivotTable Count select cheyyachu. Source data type correct chesi refresh, then Value Field Settings > Sum.

10.2 Group dates

Place Order Date in Rows -> right-click a date -> Group -> select Years/Quarters/Months. If grouping fails, blanks or invalid text dates check cheyyi.

10.3 Show Values As

- % of Grand Total: contribution.
- % Difference From: growth comparison.
- Running Total In: cumulative trend.
- Rank Largest to Smallest: category/customer ranking.

10.4 Slicer and timeline

Click PivotTable -> PivotTable Analyze > Insert Slicer. Date field kosam Insert Timeline where supported. Report Connections tho one slicer multiple PivotTables ni control cheyyachu.

11. Power Query in Excel

11.1 What Power Query does

Power Query is an ETL/data preparation engine. It records repeatable Applied Steps and reloads clean output when refreshed.

11.2 Import and clean CSV

Step	Click/action	Why/result
1	Data > Get Data > From Text/CSV	Connect to file
2	Preview -> Transform Data	Open Query Editor
3	Use First Row as Headers	Correct field names
4	Set data types	Correct calculations and filters
5	Trim/Clean text, remove errors/blanks	Improve quality
6	Rename columns/steps	Maintainable query
7	Home > Close & Load	Load table or connection/model
8	Data > Refresh All	Re-run steps on updated source

Applied Steps	Prathi transformation M language step ga record avutundi. Step order result ni affect chestundi. Example: filter before group reduces input; data type before numeric calculation prevents errors.

11.3 Merge vs Append

Operation	Meaning	Example
Merge	Join columns using matching key(s)	Orders + Customer name
Append	Stack rows with compatible columns	Jan + Feb sales

Refresh failure	File path/name/column header change ayithe query step break avvachu. Error step select chesi message chudu; source schema stable-ga maintain cheyyi.

12. Data Model, Relationships and Power Pivot

12.1 Why multiple tables?

Orders fact table, Products and Customers dimension tables separate-ga load chesi Data Model relationships create cheyyachu. Lookup formulas tho columns duplicate cheyyalsina avasaram taggutundi.

Step	Action
1	Load each table to Data Model
2	Data > Relationships (or Power Pivot Diagram View)
3	Relate dimension unique key to fact foreign key
4	Create PivotTable using workbook Data Model
5	Use fields from related tables

12.2 Measure idea

```
Total Sales := SUM(Orders[Sales])
Profit Margin := DIVIDE(SUM(Orders[Profit]), SUM(Orders[Sales]), 0)
```

A measure calculates in current PivotTable filter context. Same measure region/category selections batti new result istundi.

13. Excel Dashboard Project

13.1 Business requirements

- KPIs: Total Sales, Total Profit, Profit Margin, Orders.
- Trend: Monthly Sales and Profit.
- Comparison: Region and Category.
- Filters: Year, Region and Segment.
- Detail: Top 10 customers/products.

13.2 Build workflow

Stage	Steps	Output/reason
Raw	Keep source untouched as Table	Audit trail
Clean	Power Query types, nulls, duplicates	Repeatable dataset
Model	Relationships/measures or clean flat table	Correct calculations
Analyze	PivotTables for KPIs/trends	Reliable summaries
Visualize	PivotCharts/cards with slicers	Interactive dashboard
Validate	Compare totals and test filters	Trust

13.3 KPI formulas

```
Total Sales: =SUM(Orders[Sales])
Total Profit: =SUM(Orders[Profit])
Profit Margin: =IFERROR(Total Profit/Total Sales,0)
Order Count: =COUNTA(UNIQUE(Orders[Order ID]))
```

Design	One screen lo top KPIs, middle trend, bottom breakdown. Same color meaning maintain cheyyi. Important question ki answer ivvani visual remove cheyyi.

14. Excel Interview Questions with Model Answers

Q: 1. Workbook vs worksheet?

Answer: A workbook is the Excel file; a worksheet is one grid/tab inside it.

Q: 2. Relative vs absolute reference?

Answer: Relative references move when copied. Absolute references such as $A$1 stay fixed.

Q: 3. What is a mixed reference?

Answer: A$1 fixes the row only; $A1 fixes the column only.

Q: 4. COUNT vs COUNTA?

Answer: COUNT counts numeric cells. COUNTA counts non-empty cells.

Q: 5. SUMIF vs SUMIFS?

Answer: SUMIF handles one condition. SUMIFS handles multiple conditions and places the sum range first.

Q: 6. IFERROR?

Answer: Returns a fallback when an expression raises an Excel error; use it after understanding the expected failure.

Q: 7. XLOOKUP vs VLOOKUP?

Answer: XLOOKUP separates lookup and return arrays, can look in either direction and provides a not-found result. VLOOKUP needs the lookup column first and uses a column index.

Q: 8. INDEX MATCH?

Answer: MATCH finds a position and INDEX returns the value at that position, enabling flexible lookups.

Q: 9. Exact vs approximate lookup?

Answer: Exact requires the same key. Approximate matches a boundary/nearest rule and needs carefully ordered thresholds.

Q: 10. What is an Excel Table?

Answer: A structured dynamic range with headers, filters, calculated columns and structured references.

Q: 11. Why use structured references?

Answer: They are readable and automatically expand with table rows/columns.

Q: 12. What is a PivotTable?

Answer: An interactive tool that groups and aggregates tabular data to show comparisons, patterns and trends.

Q: 13. Why does Pivot show Count instead of Sum?

Answer: The source field is likely text or mixed type, so Excel treats it as non-numeric.

Q: 14. How do you refresh a PivotTable?

Answer: Right-click it and choose Refresh, or use Data > Refresh All for connected objects.

Q: 15. Slicer?

Answer: A visual filter for a Table or PivotTable; report connections can link one slicer to compatible PivotTables.

Q: 16. What is Power Query?

Answer: A repeatable data connection and transformation engine used for ETL/data preparation in Excel and other Microsoft products.

Q: 17. Merge vs append in Power Query?

Answer: Merge joins columns based on matching keys. Append stacks rows from tables.

Q: 18. What are Applied Steps?

Answer: The ordered sequence of Power Query transformations that re-runs during refresh.

Q: 19. Remove duplicates safely?

Answer: Define the business key, keep a backup/query step, remove by selected columns and validate row counts.

Q: 20. Conditional formatting?

Answer: Rules that change format based on values or formulas to highlight patterns and exceptions.

Q: 21. Data validation?

Answer: Rules restricting or guiding cell input, such as dropdown lists and date/number limits.

Q: 22. What causes #N/A?

Answer: A lookup could not find a required match, or NA was deliberately returned.

Q: 23. What causes #VALUE!?

Answer: A formula received an incompatible type or argument, such as text in a numeric operation.

Q: 24. What causes #REF!?

Answer: A formula references a deleted or invalid cell/range.

Q: 25. What causes #SPILL!?

Answer: A dynamic array output range is blocked or cannot spill into its intended cells.

Q: 26. How do you clean text numbers?

Answer: Fix spaces/separators, use VALUE/NUMBERVALUE or Power Query data types, then verify totals.

Q: 27. How do you find duplicates?

Answer: Conditional Formatting duplicate rule, COUNTIF, PivotTable counts or Power Query grouping, depending on the definition.

Q: 28. How do you create a dropdown?

Answer: Data > Data Validation > Allow List, then reference a clean source list.

Q: 29. Chart vs PivotChart?

Answer: A normal chart uses ranges/tables. A PivotChart stays linked to PivotTable fields and filters.

Q: 30. When use line chart?

Answer: For ordered time trends with a continuous sense of progression.

Q: 31. Why avoid pie with many categories?

Answer: Angles and small slices are difficult to compare; a sorted bar chart is clearer.

Q: 32. How do you make a dashboard dynamic?

Answer: Use Tables, PivotTables, slicers, timelines, dynamic arrays or query/model refresh.

Q: 33. What is a named range?

Answer: A readable name referring to a cell/range/formula, improving maintainability.

Q: 34. What is Goal Seek?

Answer: A what-if tool that changes one input to reach a target formula result.

Q: 35. What is a Data Model?

Answer: A relational model inside the workbook that connects multiple tables and supports measures/Power Pivot analysis.

Q: 36. Calculated column vs measure?

Answer: A calculated column computes per row and stores results; a measure calculates at query time under filter context.

Q: 37. How do you protect data quality?

Answer: Use Tables, validation, consistent types, controlled inputs, query steps, checks and reconciliation totals.

Q: 38. How do you handle a large Excel file?

Answer: Reduce volatile formulas/formatting, use Tables/Power Query/Data Model, avoid whole-column heavy arrays and split raw from reporting layers.

Q: 39. Volatile function?

Answer: A function that recalculates frequently, such as NOW, TODAY, RAND, OFFSET or INDIRECT in many contexts, potentially affecting performance.

Q: 40. Your Excel analysis process?

Answer: Understand requirement, preserve raw data, clean and validate, calculate, summarize, visualize, reconcile totals and document refresh steps.

15. Excel Practical Interview Test

Task	What interviewer checks
Clean messy sales file	Types, blanks, duplicates, auditability
Lookup product category	Exact match and missing-key handling
Region-wise monthly sales	Pivot grouping and numeric formatting
Top 10 customers	Sort/filter or Pivot value filter
Interactive dashboard	Layout, slicers, chart choice and validation
Explain wrong total	Text numbers, filters, hidden rows, duplicate grain

Test order	First raw copy -> convert to Table -> inspect types/blanks -> clean -> calculations -> Pivot -> chart -> validate grand total. Steps explain chesthu work cheyyi.

16. 30-Day Excel Plan

Days	Topics	Output
1-3	Interface, tables, formatting, references	Clean data table
4-7	Math, IF, SUMIFS, COUNTIFS	Formula practice file
8-10	XLOOKUP, INDEX MATCH	Lookup workbook
11-13	Text/date functions	Cleaning worksheet
14-15	Sort/filter/validation/formatting	Quality-controlled sheet
16-19	PivotTables, grouping, show values	Analysis report
20-22	Charts, slicers, dashboard layout	Interactive dashboard
23-26	Power Query, merge, append	Refreshable pipeline
27	Data Model and measures	Multi-table Pivot
28	Final project	Portfolio workbook
29-30	Interview and mock practical	40 answers + timed test

Official References and Version Note

Software interfaces update avutayi. Ee guide concepts and click paths official documentation tho verify chesanu. Option exact place kanipinchakapothe ribbon search/help use cheyyi.

Official source	URL
Excel help and learning	https://support.microsoft.com/en-us/excel
Create PivotTables	https://support.microsoft.com/en-us/office/create-a-pivottable-to-analyze-worksheet-data
Power Query and Power Pivot	https://support.microsoft.com/en-us/excel/learn-to-use-power-query-and-power-pivot-in-excel
Excel slicers	https://support.microsoft.com/en-us/office/use-slicers-to-filter-data
Excel table relationships	https://support.microsoft.com/en-us/office/create-a-relationship-between-tables-in-excel